This is a dense paragraph of text mixed with inline math such as $e^{i\pi} + 1 = 0$, $\sqrt{2}$, $\frac{a^2+b^2}{c^2}$, and $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ to test how prose and formulas coexist in the exported PDF.

$$f(x) = 3$$

We continue with limits and convergence language, where (x_n) converges to L if for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n - L| < \varepsilon$ whenever $n > N$.

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

Long subscripts and conditional probability should not break layout or spacing.

$$\mathbb{P}(X \leq t \mid \mathcal{F}_{\tau \wedge \sigma_{\text{very_long_stopping_time}}}) = p(t)$$

Inline calculus like $\int_0^1 x^2 \, dx = \frac{1}{3}$ appears frequently in text, while displayed equations remain simple and isolated.

$$\int_0^\infty e^{-ax} \cos(bx) \, dx = \frac{a}{a^2 + b^2}$$

Linear algebra often shows up in compact form inside PDFs.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 2 & 1 & 0 \end{pmatrix}$$

Norms, eigenvalues, and inequalities stress fonts and symbol alignment.

$$\|x\|_2 = \sqrt{x^\top x}$$

Piecewise-defined functions are common and should render cleanly.

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Finally, a long but still single-statement expression to test horizontal spacing and line wrapping behavior.

$$\prod_{j=1}^{m} \left(1 + \frac{1}{j}\right) = m + 1$$